

DRMD Best Practice (v1.0)

16.3 Instrument Vendor Checklist

A. Material Identification & Integration

- ☐ Library Mapping: Map material to internal libraries using stable materialIdentifiers.
- ☐ Deterministic Linking: Use identifiers as canonical keys for multi-material documents.
- ☐ Traceability Verification: Confirm document refers to a clearly identified material.

B. Property Data Ingestion & Calibration

- ☐ Certified Value Prioritization: Filter blocks by @isCertified to ensure only certified values are used for calibration.
- ☐ Robust Analyte Mapping: Map measured results using propertyIdentifier (CAS, element symbols).
- ☐ Metrological Ingestion: Reliably parse numerical values, units, and uncertainties.
- ☐ Uncertainty Interpretation: Correctly interpret confidence levels and coverage factors.
- ☐ Property Families: Group quantities into logical output tables.

C. Workflow & Operational Safety

- ☐ Sample Size Enforcement: Display minimumSampleSize and trigger automated warnings if planned mass is low.
- ☐ Handling Alerts: Use storage and handling instructions to prevent misuse.
- ☐ Method Templates: Support automated method templates by ingesting standardized metadata.
- ☐ Operational Notes: Display root-level comments in UI logs.

D. Security, Trust & Documentation

- ☐ Trust Anchor Verification: Optionally verify digital signatures.
- ☐ Integrity Policy: For Profile C, implement full cryptographic signature validation.
- ☐ Official Representation: Provide users access to the official PDF representation.

E. Data Normalization for Instruments

- ☐ Unit Normalization: Parse D-SI unit strings into internal units safely.
- ☐ Language Selection: Apply display selection algorithm.
- ☐ Numeric High-Precision: Store values as double/decimal.